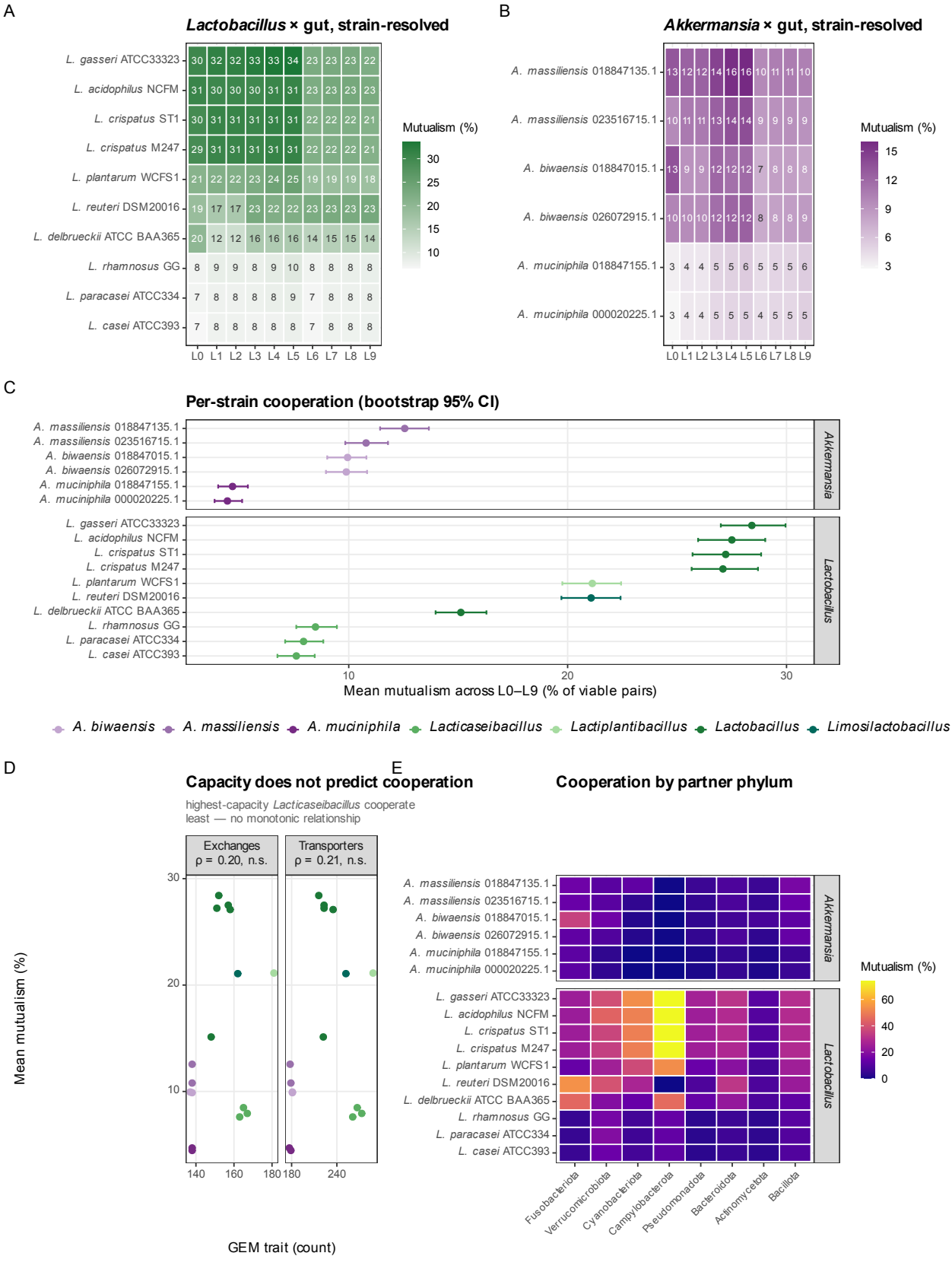

A
Lactobacillus × gut, strain-resolved
L. gasseri ATCC33323 30 32 32 33 33 34 23 23 23 22
L. acidophilus NCFM 31 30 30 30 31 31 23 23 23 23
L. crispatus ST1 30 31 31 31 31 31 22 22 22 21
L. crispatus M247 29 31 31 31 31 31 22 22 22 21
L. plantarum WCFS1 21 22 22 23 24 25 19 19 19 18
L. reuteri DSM20016 19 17 17 23 22 22 23 23 23 23
L. delbrueckii ATCC BAA365 20 12 12 16 16 16 14 15 15 14
L. rhamnosus GG 8 9 9 8 9 10 8 8 8 8
L. paracasei ATCC334 7 8 8 8 8 9 7 8 8 8
L. casei ATCC393 7 8 8 8 8 8 7 8 8 8
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9
Mutualism (%)
30
25
20
15
10
B
Akkermansia × gut, strain-resolved
A. massiliensis 018847135.1 13 12 12 14 16 16 10 11 11 10
A. massiliensis 023516715.1 10 11 11 13 14 14 9 9 9 9
A. biwaensis 018847015.1 13 9 9 12 12 12 7 8 8 8
A. biwaensis 026072915.1 10 10 10 12 12 12 8 8 8 9
A. muciniphila 018847155.1 3 4 4 5 5 6 5 5 5 6
A. muciniphila 000020225.1 3 4 4 5 5 5 4 5 5 5
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9
Mutualism (%)
15
12
9
6
3
C
Per-strain cooperation (bootstrap 95% CI)
A. massiliensis 018847135.1
A. massiliensis 023516715.1
A. biwaensis 018847015.1
A. biwaensis 026072915.1
A. muciniphila 018847155.1
A. muciniphila 000020225.1
Akkermansia
L. gasseri ATCC33323
L. acidophilus NCFM
L. crispatus ST1
L. crispatus M247
L. plantarum WCFS1
L. reuteri DSM20016
L. delbrueckii ATCC BAA365
L. rhamnosus GG
L. paracasei ATCC334
L. casei ATCC393
Lactobacillus
10
20
30
Mean mutualism across L0–L9 (% of viable pairs)
A. biwaensis
A. massiliensis
A. muciniphila
Lacticaseibacillus
Lactiplantibacillus
Lactobacillus
Limosilactobacillus
D
Capacity does not predict cooperation
highest-capacity Lacticaseibacillus cooperate least — no monotonic relationship
Exchanges ρ = 0.20, n.s.
Transporters ρ = 0.21, n.s.
Mean mutualism (%)
30
20
10
140
160
180
180
240
GEM trait (count)
E
Cooperation by partner phylum
A. massiliensis 018847135.1
A. massiliensis 023516715.1
A. biwaensis 018847015.1
A. biwaensis 026072915.1
A. muciniphila 018847155.1
A. muciniphila 000020225.1
Akkermansia
L. gasseri ATCC33323
L. acidophilus NCFM
L. crispatus ST1
L. crispatus M247
L. plantarum WCFS1
L. reuteri DSM20016
L. delbrueckii ATCC BAA365
L. rhamnosus GG
L. paracasei ATCC334
L. casei ATCC393
Lactobacillus
Fusobacteriota
Verrucomicrobiota
Cyanobacteriota
Campylobacterota
Pseudomonadota
Bacteroidota
Actinomycetota
Bacillota
Mutualism (%)
60
40
20
0